

AULA BÔNUS 14

Prof. Sérgio Altenfelder

- EQUAÇÃO DO 1º GRAU

Dso

DIREITO SIMPLES E OBJETIVO

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EQUAÇÃO DO 1º GRAU (parte 1)

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EQUAÇÃO DO PRIMEIRO GRAU

$a \cdot x + b = 0$
 $a \cdot x = -b$
 $x = -b/a$

$+$ $\rightarrow$ $-$
 $-$ $\rightarrow$ $+$
 $\times$ $\rightarrow$ $\div$
 $\div$ $\rightarrow$ $\times$

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Exemplos:

$2x + 4 = 0$
 $2x = 0 - 4$
 $2x = -4$
 $x = \frac{-4}{2}$
 $x = -2$

$2x - 4 = 0$
 $2x = 4$
 $x = \frac{4}{2}$
 $x = 2$

$-2x + 4 = 0$
 $-2x = -4$
 $x = \frac{-4}{-2}$
 $x = 2$

$-2x - 4 = 0$
 $-2x = 4$
 $x = \frac{4}{-2}$
 $x = -2$

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Exemplos:

$$2x + 4 = 6$$

$$2x = 6 - 4$$

$$2x = 2$$

$$x = \frac{2}{2}$$

$$x = 1$$

$$2x - 4 = -2x$$

$$2x + 2x = 4$$

$$4x = 4$$

$$x = \frac{4}{4}$$

$$x = 1$$

$$-2x + 4 = 4x - 8$$

$$-2x - 4x = -8 - 4$$

$$-6x = -12$$

$$x = \frac{-12}{-6}$$

$$x = 2$$

$$-2x - 4 + 4x = 6 - 3x$$

$$-2x + 4x + 3x = 6 + 4$$

$$5x = 10$$

$$x = \frac{10}{5}$$

$$x = 2$$

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Exemplos:

$$\frac{2}{3}x + \frac{4}{1} = \frac{0}{1}$$

$$2x + 12 = 0$$

$$-3$$

$$2x + 12 = 0$$

$$2x = -12$$

$$x = \frac{-12}{2} = -6$$

$$\frac{2}{3}x + \frac{2}{1} = \frac{1}{5}$$

$$10x + 30 = 3$$

$$-30$$

$$10x + 30 = 3$$

$$10x = 3 - 30$$

$$10x = -27$$

$$x = \frac{-27}{10} = -2.7$$

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Exemplos:

$$\begin{aligned} \frac{-2x}{1} + \frac{3}{4} &= \frac{2}{3}x \\ \hline -24x + 9 &= 8x \\ \hline -24x - 8x &= -9 \\ -32x &= -9 \\ x &= \frac{-9}{-32} = \frac{9}{32} \end{aligned}$$

$$\begin{aligned} \frac{3}{4}x - \frac{1}{6} &= \frac{2x}{1} + \frac{1}{1} \\ \hline 9x - 2 &= 24x + 12 \\ \hline 9x - 24x &= 12 + 2 \\ -15x &= 14 \\ x &= \frac{14}{-15} = -\frac{14}{15} \end{aligned}$$

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Exemplos:

$$\begin{aligned} \frac{x-3}{4} &= \frac{3}{2} \\ \hline 1 \cdot (x-3) &= 6 \\ \hline x-3 &= 6 \\ x &= 6+3 \\ x &= 9 \end{aligned}$$

$$\begin{aligned} \frac{x-3}{4} &\neq \frac{3}{2} \\ \hline 2 \cdot (x-3) &= 4 \cdot 3 \\ 2x-6 &= 12 \\ 2x &= 12+6 \\ x &= \frac{18}{2} = 9 \end{aligned}$$

$$\begin{aligned} \frac{2x+3}{5} &\neq \frac{x-5}{2} \\ \hline 5 \cdot (x-5) &= 2 \cdot (2x+3) \\ 5x-25 &= 4x+6 \\ 5x-4x &= 6+25 \\ x &= 31 \end{aligned}$$

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EQUAÇÃO DO 1º GRAU

(parte 2)

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EXERCÍCIO DE FIXAÇÃO

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1. Calcule o valor de x:

a) $4x + 4 = 0$

$$4x = -4$$

$$x = \frac{-4}{4}$$

$$x = -1 //$$

b) $\frac{2}{3}x = 40$

$$\frac{2x}{3} \times \frac{3}{1} = 40 \times \frac{3}{1}$$

$$2x = 120$$

$$x = \frac{120}{2} = 60 //$$

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c) $\frac{1}{3}x + 30 = 40$

$$\frac{x}{3} = 40 - 30$$

$$\frac{x}{3} \times \frac{3}{1} = 10 \times \frac{3}{1}$$

$$x = 30 //$$

d) $\frac{2}{3} \cdot \frac{1}{5} \cdot \frac{4}{6}x = 40$

$$\frac{8x}{90} \times \frac{90}{1} = 40 \times \frac{90}{1}$$

$$8x = 40 \cdot 90$$

$$x = \frac{3600}{8}$$

$$x = 450$$

12

$$e) \frac{5-x}{6} \neq \frac{x}{4}$$

$$6 \cdot x = 4 \cdot (5-x)$$

$$6x = 20 - 4x$$

$$6x + 4x = 20$$

$$10x = 20$$

$$x = \frac{20}{10} = 2 //$$

$$f) \frac{3+x}{2} \neq \frac{x}{3}$$

$$3 \cdot (3+x) = 2 \cdot x$$

$$9 + 3x = 2x$$

$$3x - 2x = -9$$

$$x = -9 //$$

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$$g) 5x - 3x + 11 = 0$$

$$2x = -11$$

$$x = \frac{-11}{2}$$

$$h) 6x + 8 = 2x + 4$$

$$6x - 2x = 4 - 8$$

$$4x = -4$$

$$x = \frac{-4}{4} = -1 //$$

14

$$\text{i) } 2x + 9 = 5x + 15$$

$$2x - 5x = 15 - 9$$

$$-3x = 6$$

$$x = \frac{6}{-3}$$

$$x = -2$$

$$\text{j) } 6x + 10 = 8x + 2$$

$$6x - 8x = 2 - 10$$

$$-2x = -8$$

$$x = \frac{-8}{-2}$$

$$x = 4$$

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$$\text{k) } 2 \cdot (x - 4) + 3 \cdot (x - 1) = 4$$

$$2x - 8 + 3x - 3 = 4$$

$$5x = 4 + 3 + 8$$

$$5x = 15$$

$$x = \frac{15}{5}$$

$$x = 3$$

$$\text{l) } 4 \cdot (x + 1) - 2 \cdot (x - 4) = 3 \cdot (x + 2)$$

$$4x + 4 - 2x + 8 = 3x + 6$$

$$2x + 12 = 3x + 6$$

$$2x - 3x = 6 - 12$$

$$-1x = -6 \rightarrow x = \frac{-6}{-1}$$

$$x = 6$$

$$x = 6$$

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$$m) \frac{4}{3}x + \frac{3}{2}x = \frac{34}{6}$$

$$\underline{8x + 9x = 34}$$

~~6~~

$$17x = 34$$

$$x = \frac{34}{17}$$

$$x = 2$$

$$n) \frac{3}{5}x + \frac{x}{10} = \frac{2}{15}$$

$$\underline{18x + 3x = 4}$$

~~30~~

$$21x = 4$$

$$x = \frac{4}{21}$$

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$$o) \frac{x+1}{4} - \frac{2x-3}{3} = \frac{1}{6}$$

$$\underline{3 \cdot (x+1) - 4 \cdot (2x-3) = 2}$$

~~12~~

$$\underline{3x+3} - \underline{8x+12} = 2$$

$$-5x + 15 = 2$$

$$-5x = 2 - 15$$

$$-5x = -13$$

$$x = \frac{-13}{-5}$$

$$x = \frac{13}{5}$$

18

$$p) \frac{4.(x-6)}{3} + \frac{5.(x+4)}{2} = \frac{1}{1}$$

$$\underline{8.(x-6) + 15.(x+4) = 6}$$

~~6~~

$$\underline{8x - 48} + \underline{15x + 60} = 6$$

$$23x + 12 = 6$$

$$23x = 6 - 12$$

$$23x = -6$$

$$x = \frac{-6}{23}$$

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GABARITO DAS QUESTÕES

a) - 1

b) 60

c) 30

d) 450

e) 2

f) - 9

g) -11/2

h) -1

i) -2

j) 4

k) 3

l) 6

m) 2

n) 4/21

o) 13/5

p) - 6/23

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